From Equity Pairs to Options Relative Value

Does reinforcement learning improve capital allocation?

Adam Kreitzman and Soo Xin Yang
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Abstract

We adapt equity-pairs trading to relative implied volatility in approximately 30-day, 25-delta out-of-the-money puts on related equity underlyings. First, a screening rule identifies trade opportunities. Direct policy-search reinforcement learning (RL) then chooses their entry size, using a decision tree and the cross-entropy method. We compare this policy with constant full sizing in a simulated $10 million account over January 2022–August 2025. Annualized excess Sharpe rises from 1.055 to 1.263, a gain of 0.208 with a conditional 95% interval of $[-0.103, 0.424]$. The learner admits 110 trades; constant full sizing admits 75. Both produce positive historical profit above cash, and learned sizing has the higher point estimate. This does not establish a repeatable advantage: results vary across years, the history was previously inspected, training and evaluation conventions differ, and execution is simplified.

1 Introduction

Suppose AMD and NVDA face similar news, but AMD options become unusually expensive relative to NVDA options. Selling the richer options and buying the cheaper ones expresses a view about the relative price of volatility. We hedge the stock exposure daily to reduce immediate directional risk. We also match the legs' initial vega to reduce sensitivity to a common small change in implied volatility. These hedges leave other risks. Related stocks can experience different events, and their option premiums may correctly reflect those differences.

Ning and Lee's equity-pairs framework [1] separates two decisions: constructing a spread and learning how to trade it. We use the same distinction for options. We first ask whether a screened relative-volatility rule produces useful historical trades under explicit contract and funding assumptions. We then ask: *does learning improve how capital is allocated to those opportunities?* Evidence that a signal is profitable does not, by itself, show that learning improves its allocation.

The report follows that sequence. We explain the motivating paper, define our option trade, and describe the monthly tree/CEM sizing procedure. We then compare constant and learned sizing under the same funded-account rules and examine annual results. The comparison uses the retained configuration and saved historical backtests.

We measure performance using annualized Sharpe of daily returns above the cash benchmark, profit above cash and cash-relative drawdown. Appendix B defines these measures and records the opening portfolios at the reporting boundary.

2 Motivating paper and adaptation

What the paper does

Ning and Lee [1] first construct a mean-reverting equity spread. Let $S_{1,t}$ and $S_{2,t}$ be the two stock prices at time t. Their spread is $X_t = S_{1,t} - BS_{2,t}$, where B is a hedge coefficient. They choose B by grid search to minimize empirical mean reversion time: the average time from a significant local extreme to the next mean crossing. This formation step does not fit an Ornstein–Uhlenbeck (OU) mean-reversion model.

They then use tabular Q-learning to trade the spread. The state records the direction and magnitude of recent spread moves. The available actions are to buy, hold or sell a unit, subject to the current position. The policy opens a long spread and later closes it; it does not open a short spread. The reward favors buying below and selling above the equilibrium mean. It also includes the transaction-cost term $c|A|$, where c is the cost coefficient and $|A|$ the action's magnitude. Q-values combine the immediate reward with the estimated value of subsequent decisions.

Training precedes the historical market test. Because the reward requires a true equilibrium mean, the agent trains on 10,000 one-year simulated OU paths. The market study then forms ten US pairs in 2022 and trades them in 2023. For all ten pairs, the published daily Sharpe ratios exceed the Distance Method and OU benchmarks. These daily figures are not directly comparable with our annualized options Sharpe ratios.

Criticisms that motivate our design

Three issues limit interpretation. **Short history:** one trading year offers little evidence across regimes. **Costs and scale:** the reward includes a cost term, but the numerical implementation and reconciliation to net performance are not clearly documented. That does not show that costs were omitted. Higher turnover increases total costs, while larger or faster orders can increase per-unit slippage and market impact. A constant cost coefficient therefore does not establish scalability. **Simulation transfer:** spread formation is empirical, but training assumes specified mean-reverting dynamics and a known equilibrium. Real relationships can shift; extensive synthetic experience alone does not establish transfer to markets.

What we borrow and what we change

We retain the separation of opportunity construction from trading decisions and ask whether learning improves on a simple rule. Our instrument changes from equity-price spreads to option-implied-volatility spreads. We use actual contracts, match initial vega and hedge stock delta daily. For training, we replace the known synthetic equilibrium with completed historical outcomes. We also specify simulated costs explicitly and compare multiple trading years.

The next section follows a trade from its relative-IV signal through contract selection and hedging to expiry. We then explain how a regression tree groups completed historical opportunities. Direct policy-search RL uses the cross-entropy method (CEM) to choose an entry size for each group. The comparison with constant sizing measures the additional historical value of learned allocation.

3 Our strategy and evaluation

3.1 From a volatility signal to a hedged trade

Implied volatility (IV) is the volatility that reproduces an option quote under a pricing model. Realized volatility (RV) measures stock-price variation; our RV inputs use only trailing data available at the signal date. We trade puts near 25 delta and 30 days to expiry (P25), with strikes below the stock price. At delta -0.25, a long 100-share contract initially has approximately -25 shares of stock exposure. This segment was retained after exploratory research and stays fixed across sizing policies; we do not claim it is optimal.

We first determine the trade direction. For underlyings A and B, let $IV_{A,t}$ and $IV_{B,t}$ be the selected-option IVs on signal date t. Rank their difference, $D_t = IV_{A,t} - IV_{B,t}$, within the 252-session history ending on that date, requiring at least 63 observations. Above the 80th percentile, sell A's relatively rich put and buy B's. At or below the 20th percentile, reverse the direction. Comparing the pair with its own history allows for persistent differences between its two IV levels.

We next check relatedness and completed profitability. At monthly formation, estimate A-on-B return beta and correlation over the preceding 126 sessions. Beta must be between 0.85 and 1.15; correlation must be at least 0.5. For the proposed direction, full-size standalone simulations must also show positive average net profit per initial vega in both the preceding 91 and 365 calendar days, with at least 21 and 63 completed outcomes, respectively. These simulations use training costs, may overlap, and need not have been executed by the funded account. An unfinished trade's eventual profit cannot inform the current decision.

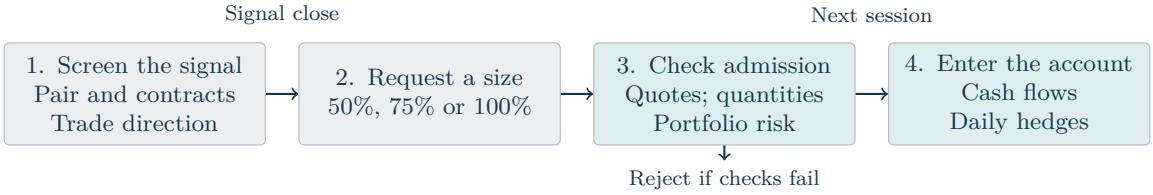


Figure 1: A signal becomes a proposal before it becomes a funded trade. Every policy sees the same proposals; requested size and existing holdings affect admission.

Before admission, we choose quantities to approximately match initial vega: the option-value sensitivity to a one-percentage-point IV increase. Table 1 illustrates the entry positions at delta -0.25 per share and 100 shares per contract.

Position	Contracts	Vega per IV point	Initial stock hedge
Buy A puts	+100	+\$450	Buy 2,500 shares
Sell B puts	−150	−\$450	Short 3,750 shares

Table 1: Illustrative vega matching: A contracts have \$4.50 vega each; B contracts have \$3.00.

After entry, hold each option to its own expiry and adjust stock delta hedges daily. Net profit combines premiums, settlement payoffs and realized hedge cash flows, after costs and financing. The IV rank is a hypothesis about relative option richness. Neither initial vega matching nor temporary IV convergence guarantees a profitable expiry outcome.

3.2 Simulating trades in a funded portfolio

Each policy starts with one $10 million account. Historical quotes, prices, rates and dividends come through WRDS. We inherit a 345-pair roster and formation calendar from earlier research. Available prepared candidates cover 125 pairs and 161 equity and ETF underlyings. After screening, 46 pairs generate 608 proposals over 2018–2025, including 312 in the later period. Thus, prior research and data availability condition the universe; it was not independently selected for this comparison.

The signal selects contracts at one close; entry occurs, if eligible, at the next session's close. We round to whole contracts and require initial vega mismatch of at most 10%. We then check the proposed trade alongside existing positions. All positions share cash and capital, and stock hedges in the same name are netted. An earlier trade can therefore prevent a new entry. Every policy receives the same proposals, quote checks and admission rules. Appendix A gives the contract and risk checks.

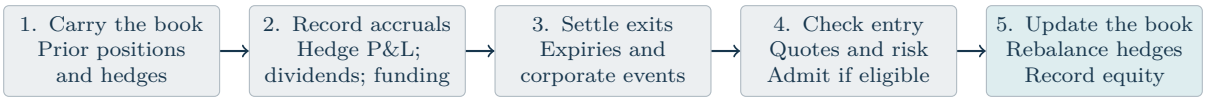


Figure 2: The daily account sequence. Existing trades remain in the shared book as new proposals arrive; the next session begins with the updated holdings.

Execution charges apply when the book trades. Options fill at midpoint, with a charge equivalent to 0.10 IV percentage point of traded vega plus $0.03 per contract. Stock hedge changes cost one basis point of turnover. On day t, let $\mathcal{F}_t$ denote paid option fills, $\Delta n_{j,t}$ the contracts traded in option j, and $V_{j,t}$ one contract's dollar vega per IV point. For stock s, $P_{s,t}$ is price and $\Delta h_{s,t}$ the change in net hedge shares. These charges give dollar execution cost

$$C_t^{\text{exec}} = \sum_{j \in \mathcal{F}_t} |\Delta n_{j,t}|(0.10 V_{j,t} + 0.03) + 0.0001 \sum_s P_{s,t} |\Delta h_{s,t}|.$$

Expiry settlement carries no option execution charge.

Funding accrues between sessions. Short stock incurs 0.3% annual borrow; negative cash incurs a 1% surcharge. Let B_{t-1} be prior cash, H_{t-1}^- prior short-stock notional, and $\Delta\tau_t$ elapsed calendar days divided by 365. Dollar funding charges are

$$C_t^{\text{fund}} = [0.003 H_{t-1}^- + 0.01 \max(-B_{t-1}, 0)]\Delta\tau_t.$$

Cash also earns or pays interest, $I_t = r_{t-1} B_{t-1} \Delta\tau_t$, where r_{t-1} is the prior annual benchmark rate as a decimal. Negative cash pays both this rate and the surcharge. Appendix A summarizes corporate-event exits, missing-quote handling and exercise approximations.

Finally, we compare account equity with an all-cash benchmark. This isolates trading value after premiums, settlement, hedges, dividends, costs and financing. Unused cash earns interest in either case, so total account growth alone does not measure trading performance.

3.3 Training, selection and testing

We select a model specification once, then refit its tree and size map monthly. Each fit uses historical examples from the preceding 18 months and controls the next month's entry decisions. Figure 3 separates these two decisions. Training and application use successive slices of the same market history.

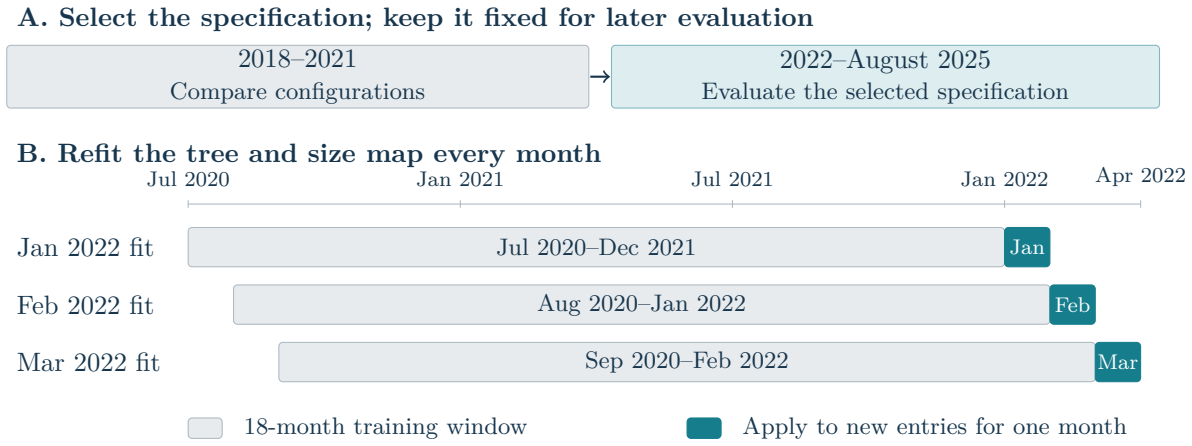


Figure 3: Two decisions, one historical dataset. The upper panel selects the 27-input, 18-month, 50/75/100% specification using earlier performance. The lower panel illustrates monthly refitting during evaluation. A training example is included only if its signal date falls within the window and its trade outcome is completed before that month's fit.

For specification selection, we compare 33 existing configurations on annualized daily excess Sharpe over 2018–2021. They vary the training history, features and sizing actions; candidate identifiers break score ties. The winner uses 27 inputs, an 18-month window and 50/75/100% actions. Its selection score is 0.575 in a $345 million research account. We then replay that configuration at $10 million, using the account assumptions in Appendix A.

For each monthly fit, an example must meet three conditions: it passed the entry screens on its signal date, has a valid simulated outcome, and finished before the fit. Its signal date must also fall within the preceding 18 months. An example need not have been admitted to the funded account. Tree fitting and CEM use this same sample; the profitability screens draw on the wider candidate history.

The later comparison runs from 3 January 2022 to 29 August 2025. The selected specification stays fixed, but monthly refitting continues. For example, a completed March trade can inform an April fit; an unfinished outcome cannot. New signals stop on 30 June 2025. Each account carries forward its own cash and holdings from 2018. The 2022 boundary separates reporting periods without resetting the accounts.

Two qualifications apply. Later history was already examined during strategy development. Monthly fits respect time order, but the later period is not independent of earlier contract, screen and universe choices. Training also charges $0.65 commission per option contract, while final evaluation charges $0.03 for every policy. We therefore interpret the comparison as historical evidence.

3.4 Learning an entry size with direct policy-search RL

The opportunity rule has already fixed the pair, direction and contracts. Direct policy-search RL now chooses the entry size: 50%, 75% or 100% of a reference budget of $450 vega per IV point on each leg. The budget scales with current account equity relative to initial equity. To learn from differences in historical rewards, we first group opportunities with a tree, then choose a common size for each group (Figure 4).

Each monthly regression tree uses 27 inputs: 15 numerical characteristics and 12 availability flags. These describe relative IV, trailing realized variance, relatedness, maturity, the CBOE Volatility Index (VIX), costs and completed profitability; Appendix A.1 defines them. The tree predicts historical full-size profit above cash and chooses splits to reduce squared prediction error. Its up to eight leaves form the states. CEM then searches for one size to assign to each state.

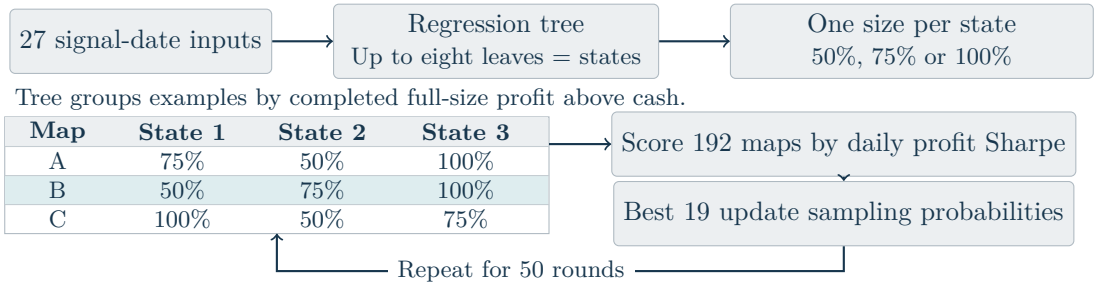


Figure 4: Tree partitioning followed by CEM sizing search. The three-state assignments illustrate candidate maps; actual trees have up to eight states. The retained map applies to the next month's new proposals.

How CEM scores a size map

To score a map, CEM combines completed trades' daily profit paths. Each example has a separate path at each size, with its own rounded contracts, standalone cash and hedge flows. These are not scaled full-size outcomes. CEM selects paths using the map, sums them by trading day, and calculates Sharpe.

Map $\boldsymbol{m}$ assigns $a_\ell \in \{0.50, 0.75, 1.00\}$ to state ℓ. Example i has inputs x_i and state $\ell(x_i)$; $p_{i,d}(a)$ is its profit above cash on day d at size a. A common stored normalization cancels from the score. Combined daily profit and training Sharpe are

$$g_d(\boldsymbol{m}) = \sum_i p_{i,d}\Big(a_{\ell(x_i)}\Big), \qquad \widehat{S}_{\text{train}}(\boldsymbol{m}) = \sqrt{252}\,\frac{\overline{g(\boldsymbol{m})}}{s_{g(\boldsymbol{m})}}.$$

Here g_d is combined daily profit, and the ratio uses its sample mean and standard deviation. The factor $\sqrt{252}$ annualizes using 252 trading days. The calendar retains overlapping trades and zero-profit sessions. This training score uses standalone dollar profits. The funded-account Sharpe instead uses percentage returns after joint admission checks.

CEM searches directly for a policy with a high Sharpe reward; it does not estimate Q-values as in the motivating paper. In each of 50 rounds, it scores 192 maps and updates size probabilities toward the best 19. Constant maps and the best map found so far remain anchors. The retained map supplies each new proposal's requested size, which still must pass account admission. This search does not establish a global optimum.

4 Benchmark and learned-sizing results

4.1 Does learned entry sizing improve the benchmark?

We first compare policies receiving the same proposals, directions and contracts. Figure 5 plots their profit above cash. Most of the learner's lead develops in 2022; both the learner and constant-full policy lose ground during 2024.

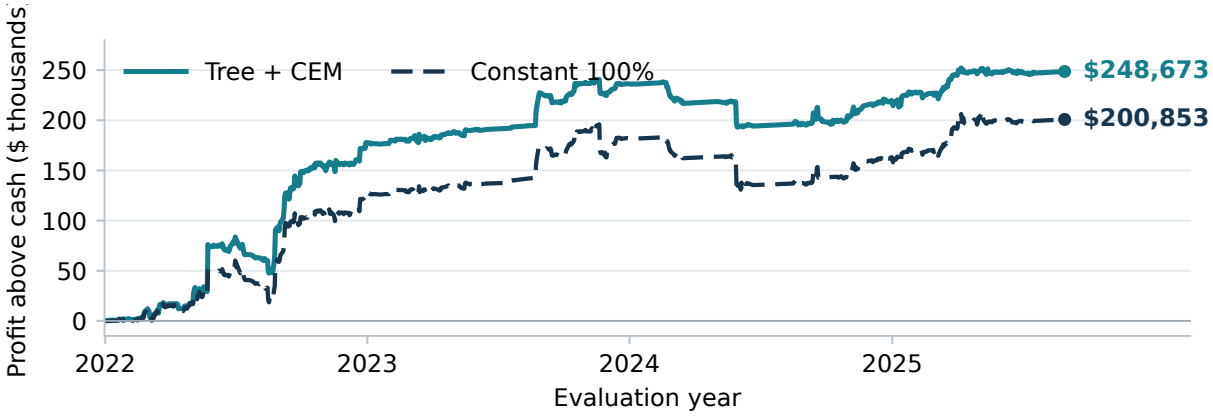


Figure 5: Cumulative profit above cash, January 2022–August 2025, rebased at 31 December 2021. Accounts retain carried positions. Endpoints match Table 2.

Compared with full sizing, RL raises later Sharpe by 0.208 and profit above cash by approximately $47,800. It also admits more entries: 110 versus 75. We then check rank and regression sizing, along with alternative linear and additive RL policies that use the same training Sharpe objective.

Policy	Earlier Sharpe	Later Sharpe	Later entries	Profit above cash ($)
Constant 50%	0.437	1.007	151	204,913
Constant 75%	0.449	0.944	104	195,021
Constant 100%	0.478	1.055	75	200,853
RL: tree + CEM	**0.573**	**1.263**	110	248,673
Spread-rank sizing	0.464	0.815	121	164,770
Ridge-score sizing	0.421	0.890	94	179,745
Direct linear policy	0.267	0.873	96	171,107
Direct additive policy	0.302	1.013	97	179,598

Table 2: Earlier: 2018–2021 replay at $10 million; later: January 2022–August 2025. The lower four rows are follow-up checks, not original selection candidates. Earlier scores are $10 million replays, not configuration-selection scores.

Among these policies, tree/CEM has the highest funded Sharpe in both periods. The alternatives change the sizing pipeline as a whole, not only the tree architecture. To assess uncertainty in the later gain over full sizing, we use a paired circular-block bootstrap with 126-session blocks and 1,000 replications. Its interval, $[-0.103, 0.424]$, includes zero and does not adjust for research search.

4.2 Why does the admitted portfolio matter?

To understand the result, we first sum every proposed standalone trade path at evaluation costs. Later Sharpe is then 0.606 for tree/CEM and 0.726 for the additive policy. Next, retain only the trades each policy's funded account admitted. The ranking reverses: 1.268 versus 1.027, close to the funded-account results of 1.263 versus 1.013. Smaller sizes can pass pair stress limits and leave room for later entries. The diagnostic therefore shows an interaction between sizing and admission. It does not isolate predictive sizing or prove that the learner anticipated the constraints.

The learner's annualized total account return is 4.667%, but annualized growth relative to cash is 0.530%. Most total growth therefore comes from interest; trading value is measured above cash. Average reserve is 2.44% and peak reserve is 15.77%. Cash-relative drawdown is 0.495%, versus 0.646% at full sizing.

4.3 Variation across years and pairs

We next examine whether the gain is spread across years and pairs. Table 3 shows uneven annual performance. In 2024, the account grows in total but loses relative to cash.

Year	Account return (%)	Above cash (pp)	Excess Sharpe	Entries
2022	3.470	1.673	2.458	66
2023	5.288	0.432	1.492	19
2024	5.104	−0.283	−0.907	11
2025 (partial)	3.207	0.169	1.092	14

Table 3: Tree/CEM annual funded results; 2025 ends on 29 August. Account and above-cash returns are unannualized for each displayed period, including partial 2025. Above-cash figures are return differences in percentage points; Sharpe is annualized.

Profit is concentrated by pair and year. SNAP/UBER, GAP/TJX and AMD/MU contribute 66.9% of later net pair trading profit. Across all pairs, 2022 accounts for 83.1% of that later profit. These percentages exclude shared financing. They indicate concentration, rather than broad, repeated success across pairs.

5 Conclusion and extensions

The retained tree/CEM RL policy improves the historical funded point estimate over constant exposure and the tested alternative sizing pipelines. That benefit combines exposure choices with portfolio admission. The uncertainty interval, previously inspected history, inherited universe and simplified execution limit what we can conclude about repeatability and live profitability.

A natural extension is to allocate across several independently researched signals. First, freeze each signal's trade proposals. Then let RL choose exposures using current inventory, available capital, shared underlying risks and completed signal performance. Train against the actual funded ledger and compare with fixed allocations. This would test whether learning can combine opportunities into a portfolio, beyond sizing one retained signal.

A Signal and learning specification

Contract selection proceeds in two steps. In the historical anchor selection, choose the expiry closest to 30 days, then the put nearest absolute delta 0.25 within 0.075. Break ties using narrower spreads, higher open interest and lower contract identifier. At entry, require standard 100-share contracts, positive noncrossed quotes, open interest of at least 100, width of at most 25% of midpoint and 7–55 days remaining. For the signal, IV rank includes the signal close; relatedness is fixed monthly from the prior 126 sessions. Profitability memory is dated by completion, not entry.

Admission first evaluates each name's option-plus-hedge loss under nine combinations of spot shocks ($-20/0/+20\%$) and IV shocks ($-10/0/+10$ percentage points). Sum each name's worst loss, without credit for pair correlation. Stress loss may not exceed 5% of budget. Reserve, defined as gross option premium plus half netted hedge notional, is capped at 80%; hedge notional is capped at twice budget. Gross vega is limited to \$20,000 per IV point per \$1 million. Global checks use account equity; pair checks use \$1 million scaled by current/initial account equity. Process proposals in pair-identifier order. These are research limits, not broker margin requirements.

After entry, corporate events or unsupported declared distributions cause affected pairs to exit. Missing entry quotes reject a proposal. For held positions, marks may instead use last-known IV with current spot, rates and known dividends. Expiry is cash-settled using a European valuation approximation. Market impact, early assignment and forced liquidation are not fully modeled.

The tree has up to eight leaves. Its minimum leaf count is one eighth of the training sample, rounded down and bounded between 8 and 63. With fewer than 16 examples, use full sizing. Each month, CEM starts with uniform probabilities. For each update, mix 30% current probability with 70% elite frequency, multiply by 0.97, then distribute an additional 0.03 equally across the three actions. The random seed is 2026.

A.1 Inputs available at the signal date

The table defines the 15 numerical inputs. Orient proposed-direction quantities as long leg minus short leg, and express IV and volatility levels as decimal annualized values. To calculate trailing realized variance, take mean squared daily simple returns over the latest 20 listed trading sessions and multiply by 252. Require at least 10 observations and include the signal-close return. Returns generally exclude dividends; adjusted-close percentage changes provide the fallback series.

Input	Definition
Mean IV	Mean of the two selected-contract IVs.
Oriented IV gap	Long-minus-short IV divided by mean IV.
Rank strength	Twice the absolute distance of the IV-gap percentile from 0.5.
Oriented variance-premium gap	Long-minus-short difference in IV squared minus trailing realized variance, divided by mean IV squared.
Mean RV	Mean of the square roots of the two trailing realized variances.
Beta; correlation	Two inputs: 126-session A-on-B beta and correlation, fixed at monthly formation.
Mean expiry; expiry gap	Two inputs: mean calendar days to expiry and long-minus-short expiry days.
Commission in IV points	Sum across legs of \$0.65 divided by each contract's dollar vega per IV point.
VIX	Most recent strictly prior VIX close divided by 100.
91-day profit; count	Two inputs: completed full-size profit per initial vega averaged over the trailing 91 calendar days and the number of completed outcomes. The profit input is divided by 100 as a fixed scale.
365-day profit; count	Two inputs: the same measurements over 365 calendar days.

The first three inputs are core fields. Add a binary availability flag for each of the other 12, giving 27 inputs in total. Set a nonfinite optional value and its flag to zero; an observed value has flag one. Profit features use only outcomes completed before the signal date. Future returns and profits enter the outcomes, not the inputs.

B Performance definitions and evidence

We first construct the cash benchmark. Let N_t be account equity and K_t all-cash benchmark wealth at close t; both start at \$10 million in January 2018. The benchmark compounds the previous available WRDS zero-curve rate nearest 30 days over calendar days, using a 365-day year. We then subtract its daily return from the account's daily return. This gives daily excess return e_t and annualized Sharpe

$$e_t = \left(\frac{N_t}{N_{t-1}} - 1\right) - \left(\frac{K_t}{K_{t-1}} - 1\right), \qquad \widehat{S} = \sqrt{252}\,\bar{e}/s_e.$$

Here $\bar{e}$ and s_e are the sample mean and standard deviation of daily excess returns; annualization assumes 252 trading days. For a reporting period with preceding close 0 and final close T, dollar profit above cash is the change in excess wealth: $(N_T - K_T) - (N_0 - K_0)$.

To measure growth and drawdown relative to cash, form $Q_t = N_t/K_t$. Annualized relative growth is $(Q_T/Q_0)^{1/Y} - 1$, where Y is elapsed days divided by 365.25. Cash-relative drawdown is the largest percentage decline in Q_t from its running peak within the period, including close 0.

For the annual table, account return is $N_T/N_0 - 1$. Subtract benchmark return $K_T/K_0 - 1$ to obtain above-cash percentage points. These period returns are unannualized, including partial 2025; Sharpe remains annualized.

At the 31 December 2021 reporting boundary, learner equity is \$10.568 million with four open trades. Constant-full equity is \$10.581 million with three open trades; benchmark wealth is \$10.510 million. The chart subtracts each policy's opening dollar excess wealth. It does not reset the accounts or their holdings.

To estimate the gain interval, resample learner and constant-full daily excess returns using identical indices. Build each of 1,000 samples by concatenating randomly started 126-session blocks, wrapping at the series end, and truncating to the original length. Recalculate the Sharpe difference for each sample, then take its 2.5th and 97.5th percentiles. Fitted policies stay fixed; the interval does not adjust for research selection.

Code and evidence in the submission. The accompanying package contains the main strategy implementation in `paper_code/` and the bundled inputs needed to reproduce the four principal policies. Within that module, `signals.py` defines screening and inputs, `policy.py` fits and scores sizes, `ledger.py` simulates the shared account, and `metrics.py` calculates performance. `README.md` provides setup and reproduction commands. All paths below are relative to the package's `code/` directory. The directory `reports/p25_validation_split` contains `periods.csv`, `annual.csv`, `holdout_intervals.csv` and `selection.json`. Figure 5 uses the selected account in `data/p25_validation_split_experiment.zip` and the constant-full account under `data/submission/artifacts/contracts/p25_capital_sensitivity`. The complete 125-pair roster, candidate-date ranges and proposal counts are retained in the companion `paper/pair_universe.csv`; ranges describe recorded candidates, not continuous quote coverage or executed trading dates. Inherited formation calendars and usable contracts condition the universe. Supporting comparisons and concentration use `reports/p25_strategy_audit` and `reports/p25_direct_policy`; the admission diagnostic is in `reports/p25_generalization_bridge`. Vendor data retain their institutional license restrictions.

References

[1] B. Ning and K. Lee (2024). *Advanced Statistical Arbitrage with Reinforcement Learning.* arXiv:2403.12180v1. https://arxiv.org/abs/2403.12180.